

Residentia Electrical Devices

Skill Trades 1201

Circuit Protection Devices

- Circuit breakers and fuses are designed to be the weakest link in any circuit.



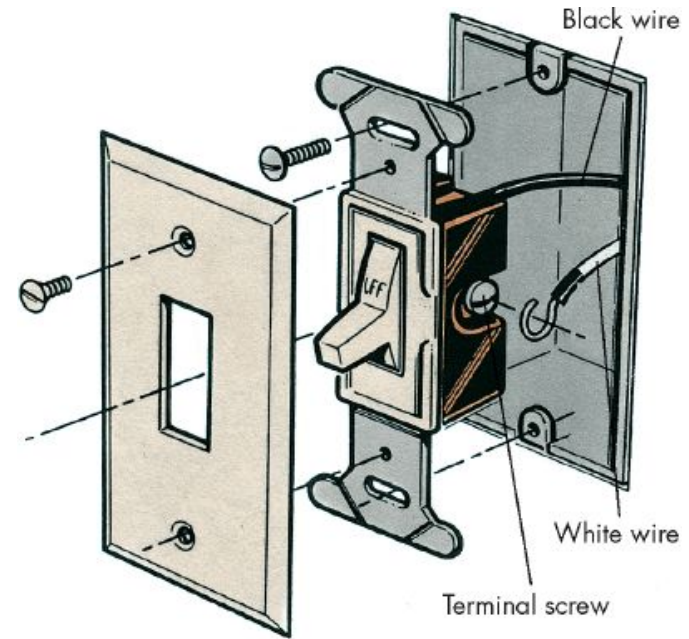
Circuit Protection Devices

- If a short or overload condition should occur, it will be the fuse or circuit breaker that will open circuit long before the wire can heat up to start a fire.



Switching Components

- Electrical devices used to control either lights or receptacles from one or more locations.
- Mounted 54 inches above the floor.
- Mounted Outside bathrooms.



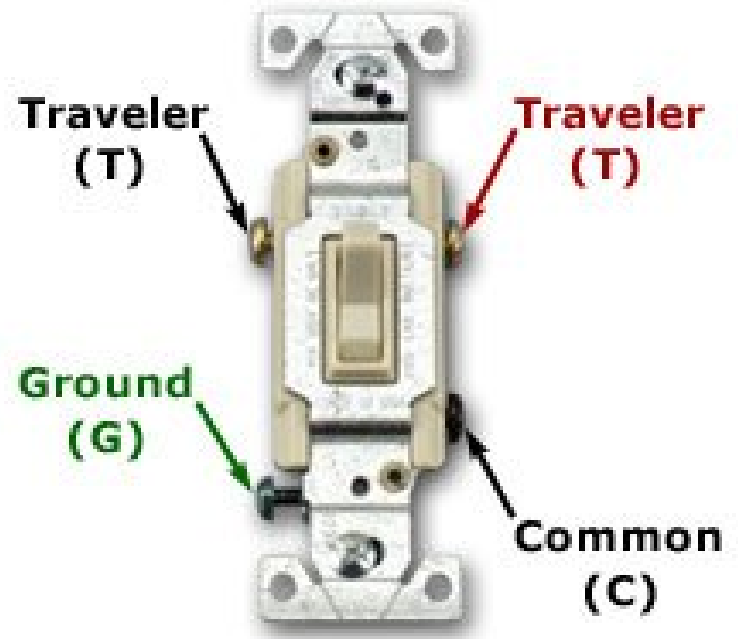
Types of Switches

- Single Pole Toggle
 - Controls a light from a single location only.



Types of Switches

- Three Way Switch
 - Controls a light from two locations.



Types of Switches

- Dimmer Switch
 - Allows you to adjust the brightness of a light.



Receptacles

- Devices used to allow appliances to plug in.
- Most widely used type of outlet used in the house.
- There should be an outlet every 12 linear feet.



Receptacles

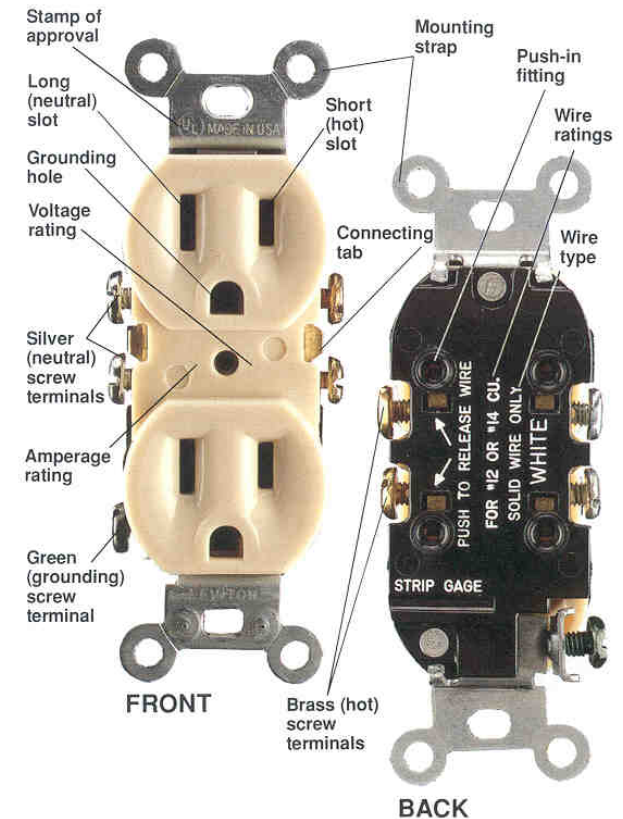
- Outlet should be mounted 12-17 inches above the floor.
- Maximum of 12 outlets permitted per 15 Amp circuit.



15 amps, 125 volts. Three-slot grounded receptacle has two different size slots and a U-shaped hole for grounding. It is required in all new wiring installations.

Types of Receptacles

- Duplex Receptacle (120 Volt)
 - Most common outlet used in the house.
 - Black wire attaches to the brass screw. White wire to the silver screw, and bare copper to the ground screw.



Types of Receptacles

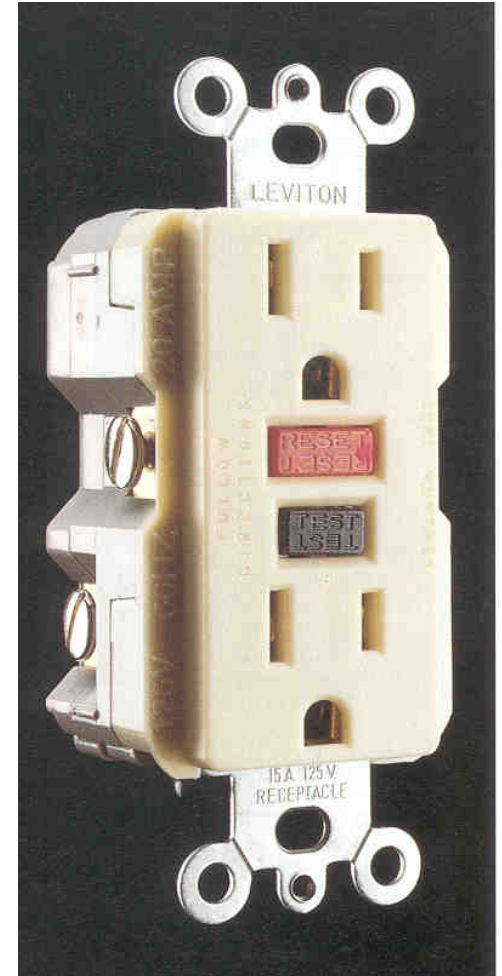
- 240 Volt Receptacles
 - Special four pronged outlet that supplies 240 volts.
 - Plug design prevents plugging in 120 volt appliances.
 - Commonly used for Range and Dryer.



30 amps, 125/250 volts. This receptacle is used for clothes dryers. It provides high-voltage current for heating coils, and 125-volt current to run lights and timers.

Types of Receptacles

- Ground Fault Circuit Interrupter (GFCI)
 - Monitors the amount of current flowing from hot to neutral.
 - If there is any imbalance, it trips the circuit.



Types of Receptacles

- Ground Fault Circuit Interrupter (GFCI)
 - Protection from severe or fatal electric shocks.
 - Bathroom Receptacles & Outside Receptacles.

